

JEE MAIN REVISION MASTER: WAVES

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Wave Equation & Velocity of Sound	5 PYQs	Wave Motion
Moderate	Standing Waves, Resonance & Doppler Effect	5 PYQs	Interference & Acoustics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Wave Equation & Velocity of Sound	$y(x,t) = A \sin(kx \mp \omega t + \phi)$, $v = \frac{\omega}{k} = \lambda f$, $v_{\text{string}} = \sqrt{\frac{T}{\mu}}$, $v_{\text{gas}} = \sqrt{\frac{\gamma RT}{M}}$
Standing Waves, Resonance & Doppler Effect	$f_{\text{open}} = \frac{nv}{2L}$; $(n=1,2,3)$, $f_{\text{closed}} = \frac{(2n-1)v}{4L}$; $(n=1,2,3)$, $f' = f \left(\frac{v \pm v_0}{v \mp v_s} \right)$, $f_{\text{beat}} = f_1 - f_2 $

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Wave Equation & Velocity of Sound (Moderate Yield — 5 PYQs)

Core Principle: Progressive traveling harmonic waves, wave vector $k = 2\pi/\lambda$, and Laplace-corrected speed of sound in gases.

■ **Caution / JEE Pitfall:** Maximum particle velocity $v_{\text{(particle, max)}} = A\omega$ is completely different from wave propagation velocity $v = f\lambda$.

■ **Speed Trick:** Wave travels in positive x-direction if signs of kx and ωt are OPPOSITE: $\sin(kx - \omega t)$.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "The angle between the particle velocity and wave velocity in a transverse wave is [except when the particle passes through the mean position..." [Ans: N/A]
- [2024 Shift] Q#4 (MCQ): "A galaxy is moving away from the Earth so that a spectral line at 600 nm is observed at 601 nm. Then, the speed of the galaxy with respect ..." [Ans: A]
- [2021 Shift] Q#1 (MCQ): "To propagate both longitudinal and transverse waves, a material must have..." [Ans: N/A]

3.2 Standing Waves, Resonance & Doppler Effect (Moderate Yield — 5 PYQs)

Core Principle: Stationary waves in strings and organ pipes, beat frequency, and Doppler frequency shift for moving sources and observers.

■ **Caution / JEE Pitfall:** A closed organ pipe produces ONLY odd harmonics (1, 3, 5...). The first overtone is the 3rd harmonic ($3v/4L$).

■ **Speed Trick:** Doppler rule: When distance decreases, apparent frequency INCREASES; when distance increases, apparent frequency DECREASES.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A source of frequency v gives 6 beats/second when sounded with a source of frequency 200 Hz. The second Harmonic of frequency $2v$..." **[Ans: B]**
- **[2024 Shift] Q#3 (MCQ):** "A motor-cyclist moving towards a huge cliff with a speed of 18kmh^{-1} , blows a horn of source frequency 325 Hz. If the speed of the sou..." **[Ans: N/A]**
- **[2024 Shift] Q#4 (MCQ):** "A galaxy is moving away from the Earth so that a spectral line at 600 nm is observed at 601 nm. Then, the speed of the galaxy with respect ..." **[Ans: A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A source of frequency v gives 6 beats/second when sounded with a source of frequency 200 Hz. The second Harmonic of frequency $2v$ of the source gives 8 beats/second when sounded with a source of frequency 420 Hz. The value of v is

- (A) 205 Hz
- (B) 206 Hz **[Correct]**
- (C) 195 Hz
- (D) 210 Hz

Correct Answer: B

Solution Analysis: For beats, the number of beats per second is equal to the difference in the two frequencies. Given that the source of frequency v gives 6 beats/second with 200Hz : $|v-200|=6$ So, $v=200+6=206\text{Hz}$ or $v=200-6=194\text{Hz}$ Now the second harmonic of the source has frequency $2v$. It gives 8 beats/second with 420Hz, so $|2v-420|=8$ This gives $2v=420+8=428$ or $2v=420-8=412$ So, $v=214\text{Hz}$ or $v=206\text{Hz}$ The common value from both conditions is $v=206\text{Hz}$ So the correct option is Option B: 206 Hz.

Benchmark Problem 2: KCET 2025 (Q#2)

The angle between the particle velocity and wave velocity in a transverse wave is [except when the particle passes through the mean position]

- (A) $(\pi / 4)$ radian
- (B) $(\pi / 2)$ radian
- (C) π radian
- (D) Zero radian

Correct Answer: None

Solution Analysis: In a transverse wave, particles oscillate in a direction perpendicular to the wave's direction of propagation. Therefore, the angle between the particle velocity and wave velocity is $(\pi / 2)$ radians, except when the particle crosses the mean position.